ASPREDICTED

Wharton CREDIBILITY LAB University of Pennsylvania

MeLiDos Project – Multi-country determinants of personal light exposure (#273407)

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Pre-registered on:
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1) Have any data been collected for this study already?

It's complicated. We have already collected some data but explain in Question 8 why readers may consider this a valid pre-registration nevertheless.

2) What's the main question being asked or hypothesis being tested in this study?

H1: Personal light exposure metrics differ significantly across sites after accounting for latitude and photoperiod.
H2: Variance in hourly melanopic EDI within participants (nested in sites) exceeds variance between sites.
H3: Hourly self-reported light exposure categories predict hourly geometric mean melanopic EDI.
H4: Hourly self-reported activity categories predict hourly geometric mean melanopic EDI.
H5: LEBA questionnaire factors correlate with selected personal light exposure metrics.
H6: Day type (weekday/weekend; free/work day), daily exercise, and sleep variables predict daily light exposure metrics.
H7: There is a ceiling (nonlinear) effect of absolute latitude and photoperiod on level-, duration-, and exposure-history-based light metrics.
H8: Duration-, exposure-history-, and level-based metrics are associated with VLSQ-8 light sensitivity scores.
H9: Timing-based metrics are associated with chronotype (MCTQ, MEQ).
H10: Personal light exposure metrics depend on age and sex.
H11: Diurnal exposure patterns differ by sex.

3) Describe the key dependent variable(s) specifying how they will be measured.

All outcomes are derived from chest-mounted wearable measurements of melanopic equivalent daylight illuminance (EDI) sampled at 10-second resolution.

Level-based metrics
• Geometric mean melanopic EDI (hourly bins; daily average)
• Geometric mean melanopic EDI during 10 brightest hours
• Geometric mean melanopic EDI during 5 darkest hours

Duration-based metrics
• Time above 1000 lx melanopic EDI
• Time above 250 lx melanopic EDI (daytime)
• Time within 1–10 lx (evening)
• Time below 1 lx (sleep)
• Longest period above 250 lx

Timing-based metrics
• First time above 250 lx
• Last time above 250 lx
• Midpoint of brightest 10 hours
• Midpoint of darkest 10 hours
• Midpoint of longest period above 250 lx

Spectrum-based
• Melanopic daylight efficacy ratio (MDER; CIE S026)

Dynamics-based
• Interdaily stability
• Intradaily variability

Exposure-history-based
• Melanopic EDI dose (lx·h)

For specific hypotheses (H2, H3, H4, H11), the dependent variable is hourly geometric mean melanopic EDI.

4) How many and which conditions will participants be assigned to?

This is an observational study. No experimental assignment occurred.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

All analyses will be conducted in R (v4.5) using tidyverse, lme4, mgcv, and LightLogR.

General:
• False discovery rate (FDR) correction within hypothesis families
• Model selection via likelihood ratio tests (χ^2) for LMMs, $\alpha = 0.05$ and $\Delta AIC \geq 2$ for GAMMs
• Diagnostics: residual inspection, homoscedasticity, normality
• If models are unstable: reduce random slopes → reduce interactions
• AR(1) structures added where residual autocorrelation is detected in GAMs

H1; Linear mixed model (LMM):
Metric ~ Site + Latitude + (1|Site:Participant)
Photoperiod included for day/night duration metrics.
H2; GAMM:
Metric ~ Site + s(Time, by=Site, bs="cc", k=12) + s(Participant, Time, bs="fs")
Variance components compared (within vs between site).
H3 & H4; LMM:
melEDI ~ Predictor + (Predictor|Site) + (1|Site:Participant)
H5; Correlation matrices + linear models:
Metric ~ LEBA + (1|Site)
H6; LMM:
Metric ~ Measure + (Measure|Site) + (1|Site:Participant)
H7; GAMM:
Metric ~ te(Latitude, Photoperiod) + s(Site, bs="re") + s(Participant, bs="re")
H8–H10; LMMs:
Metric ~ Predictor * Site + (1|Site:Participant)
H11; GAMM:
melEDI ~ Sex + s(Time, by=Sex, bs="cc", k=12)
+ s(Time, by=Site, bs="fs", k=12)
+ s(Site, bs="re") + s(Participant, bs="re")
Primary analyses use chest-level data. Analyses will be repeated using glasses-mounted data to evaluate robustness.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.
Exclusion rules:
1. Days with <80% valid wear time (excluding sleep) will be excluded from daily metric computation.
2. Non-wear periods identified via device logs will be removed.
3. Observations missing required predictors for a given model will be excluded on a per-analysis basis (complete-case within model).
4. No statistical trimming (e.g., ±3 SD) will be applied to melanopic EDI values, but values above $1.2*10^5$ will be set to NA
5. Hours with less thant 50% will be excluded

All exclusions will be reported transparently.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.
All available participants in the MeLiDos repository meeting inclusion criteria will be included.
Target per site: 15–30 participants; convenience sample; size based on power calculation (>80%)

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)
All data have already been collected as part of the multinational MeLiDos Project and are publicly available (https://github.com/MeLiDosProject)
Although the dataset has been accessed and descriptively inspected (e.g., variable structure, completeness), none of the hypothesis-driven analyses specified below have been conducted prior to this preregistration.

To reduce researcher degrees of freedom:
• Hypotheses and outcome metrics are defined a priori and have not yet been tested, even in a subsample of participants
• Statistical models are specified in advance based on theoretical assumptions, not tested on the data
• Model selection criteria and inference thresholds are fixed before hypothesis testing.
• Multiple-testing correction procedures are pre-defined.
• Any deviations will be explicitly documented and labeled as exploratory.

This preregistration therefore documents confirmatory analyses applied to existing observational data.